INFERCELL: A COMPOSITIONAL INFERENCE FRAMEWORK FOR WHOLE-CELL MODELLING, DEMONSTRATED ON REDUCED JCVI-SYN3A

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Tom Kimpson[*1,2]
[1]School of Mathematics and Statistics, The University of Melbourne, Parkville, VIC 3010, Australia
[2]ARC Centre of Excellence for the Mathematical Analysis of Cellular Systems (MACSYS), The University of Melbourne, Parkville, VIC 3010, Australia

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ABSTRACT

Whole-cell models simulate a living cell from its molecular parts. They usually fix each parameter at one value, so a prediction does not say how much of it the data support. Inference is hard because the models mix deterministic and stochastic formalisms, the stochastic paths are latent, and no parameter is shared between the two. We present INFERCELL, a compositional inference framework. Parameters carry priors and sources, and the framework checks each composition and lists its declared reductions. INFERCELL supports exact stochastic and mean-field descriptions across a coupled ODE–jump boundary, with two inference routes: an exact sampler, and a cheaper mixed route that integrates out the stochastic paths. We demonstrate it on a reduced model of the minimal cell JCVI-syn3A and on M0, a two-gene slice of that model. On M0, both routes pass simulation-based calibration over 50 datasets, with small declared discrepancies. The mixed route costs a median of 6.57 CPU-hours per posterior, against 507.5–678. On the one dataset with a long exact reference, three parameters sit 0.42–0.56 standard deviations from it, and the cause is open. Posterior-to-prediction scoring on M0 covers lactate export at the expected rate, a quantity the data measure noisily through extracellular lactate. A knockdown's effect, which the data do not measure, has the right sign, though those intervals are not a clean calibration. We then carry an M0 posterior into the reduced model, with viability-conditioned uncertainty in the remaining parameters, to make forward and intervention predictions. These predictions are unscored. At the default prior width, among the 20 parameters drawn and on one posterior, about 91% $\pm$ 4% of the parameter uncertainty in lactate export comes from 15 promoter strengths that the fit cannot reach, and that share runs from 63% to 98% as their asserted prior width changes. The data are synthetic. The framework shows what a prediction rests on. We claim no all-parameter posterior, no calibrated inference on the reduced model, and no biological finding.

Keywords whole-cell model · JCVI-syn3A · Bayesian inference · hybrid simulation · uncertainty quantification · Julia

1 Introduction

A whole-cell model simulates a cell from its molecular parts. The first such model described *Mycoplasma genitalium* with 28 coupled submodels [1]. Later models covered *Escherichia coli* [2] and the minimal cell JCVI-syn3A, first well stirred [3] and then in space and time over a full cell cycle [4]. Each one joins metabolism, gene expression, growth and division in one simulation, and each reproduces measured behaviour of a real cell.

All of these models are forward simulators. Each one takes hundreds or thousands of kinetic parameters from the literature, sets each to one value, and integrates forward in time. The output is one trajectory. That trajectory is one sample from the distribution of trajectories that the parameter knowledge allows. The model does not know the width of that distribution. When a model predicts a doubling time, a reader cannot tell whether the prediction

[*]Corresponding author: tom.kimpson@unimelb.edu.au

holds for every plausible parameter set or depends on a few uncertain values. Sensitivity analyses vary parameters [2], but they do not say what the data support.

The missing error bar is a gap in the scientific reasoning. Every parameter in a whole-cell model is uncertain. A rate measured in one organism, at one temperature, in one medium, is then used in another organism under other conditions. The uncertainty is real, and it propagates through coupled nonlinear dynamics into every prediction. A framework that stores each parameter as a single number has no place to put this uncertainty. It cannot ask how uncertain a prediction is, and it cannot ask which data would reduce that uncertainty.

1.1 Why whole-cell models do not do inference

Bayesian inference answers both questions. It treats each parameter as a distribution, updates the distribution with data, and propagates the result into predictions. This approach has been argued for whole-cell models [5]. Whole-cell parameters have been estimated, but as point estimates and not as calibrated posteriors. In the DREAM8 challenge, teams recovered 30 parameters of a modified *M. genitalium* model, 15 changed and 15 unchanged, scored against that one known truth [6]. Parameter error and prediction error had a log–log R^2 of 0.57 over all submissions and only 0.04 among the top solutions, and the authors note that several parameter sets give similar system-level phenotypes. A single estimate cannot express that, and a posterior can [7]. No team used automatic differentiation, and the authors ask for automated reduced models and for data on single-cell variation over time. Apart from Bayesian balancing of metabolic parameters, we know of no whole-cell model that reports posteriors over its parameters, and we see the following reasons.

1. *No place for uncertainty.* The frameworks store parameters as floating-point values in a dictionary. The source of each value is in a spreadsheet, a flat file or a supplement, and rarely with the value in the model.
2. *Mixed formalisms.* Metabolism is a set of ordinary differential equations (ODEs). Gene expression is a stochastic jump process. The two exchange state on a fixed clock. No single likelihood and no standard sampler covers both.
3. *Latent paths.* The stochastic block has a hidden path: the exact times of each transcription and translation event. The likelihood of the data integrates over all such paths, and that integral has no closed form.
4. *Cost.* Inference needs many simulations. For a whole-cell model the number of simulations, not the cost of one, sets the budget.
5. *No shared parameters.* In the published JCVI-syn3A model, no rate constant appears in both blocks. The blocks couple only through state. The standard methods for modular inference pass information through shared parameters [8–10]. Here those methods do not apply directly.

Table 1 compares current whole-cell frameworks on four design features. Thornburg and coworkers balanced the metabolic parameters of JCVI-syn3A against each other with a Bayesian method [3]. That step used literature values for metabolism only. It did not infer parameters from dynamic data through the coupled model.

1.2 This work

We present INFERCELL, a compositional inference framework demonstrated on reduced JCVI-syn3A. We do not claim an all-parameter posterior, calibrated inference on the reduced model, or a biological finding. We claim that a framework can make inference a basic operation, so that a modeller can apply it where the data and the budget allow.

Section 2 describes the framework. Every parameter is a prior with a recorded source. Modules couple through seven kinds of typed edge. The framework checks that a composition is complete before it runs, and it lists every departure from the published model. The same gene-expression modules run as an exact stochastic process or as mean-field ODEs, and in the second form the composed model compiles to one ODE problem that automatic differentiation can pass through.

Section 3 then follows one demonstration through three steps. First comes a test system: a reduced model of JCVI-syn3A that keeps 22 metabolic reactions, 17 genes and the two-way coupling between them, and M0, a two-gene slice small enough to check (Section 3.1). Second come the posteriors: on M0, an exact sampler gives a calibrated reference, and a mixed route passes the same test at a fraction of the cost, with small declared discrepancies (Sections 3.2 to 3.4). Third come probabilistic predictions: scored against the truth on M0 (Section 3.5), then unscored on the reduced model, where an M0 posterior is combined modularly with viability-conditioned uncertainty in the remaining parameters (Section 3.6).

Our main result is that one framework carries a reduced whole-cell model from declared priors, through inference calibrated on the slice where calibration can be checked, to unscored predictions on the reduced model, with the limits of those predictions declared.

2 InferCell

INFERCELL is written in Julia [13]. It uses the SciML solvers for ODEs and jump processes [14]. This section gives the design ideas. The documentation gives the full interface (see Software and data availability). Figure 1 shows the parts and how they connect.

Table 1: **Capabilities of whole-cell modelling frameworks.** Each column asks whether the framework does this as part of its design. Notes follow the table.

	Priors	Provenance in model	Inference on dynamics	Declared reductions
M. genitalium [1]	No	Partial (a)	Fit, no posterior	Not applicable
E. coli [2]	No	Partial (b)	Fit, no posterior	Partial (c)
Vivarium [11]	No	No	No	No
JCVI-syn3A [3, 4]	Partial (d)	Partial (e)	One block (f)	Partial (g)
INFERCELL (this work)	Yes	Yes	On M0	Yes

These columns are INFERCELL's own design aims, so the comparison favours it, and the fits in the other rows used real data where ours are synthetic. (a) A separate knowledge base with references, read by the simulation [12]. (b) Annotated flat files, with some sources blank. (c) Adjustments as hard-coded tables in the fitting code. (d) Metabolic priors in the balancing input, with their means used in simulation. (e) Sources in pre-balancing flat files, and not in the balanced table. (f) Bayesian parameter balancing of metabolism on literature values, then further adjustment. (g) Patches noted in code and files. Each cell's evidence is in `paper/tables/frameworks_sources.md`, from the papers' main text and each framework's public code. We did not read the supplements, and for the 2026 JCVI-syn3A model we read the preprint.

2.1 Parameters are distributions with a source

Each parameter in INFERCELL is an object that holds a value, a prior distribution, a role and a record of its source. The record names the file, the table and the identifier the value came from, and the values that other files give for the same identifier. Each parameter also has one of four labels for how well its prior is informed:

- *balanced*: the value and width come from a balanced parameter table;
- *prior default*: the source gives no width, so the prior is wide;
- *asserted*: the source gives no uncertainty, so we set the prior;
- *not imported*: the parameter is not in the source.

The loader does not choose silently between two files that disagree. It stops and asks for a decision, and the decision is then part of the record. A user can ask the composed model which priors no measurement supports. That question has a direct answer, because the answer is in the model and not in a supplement.

2.2 Modules couple through typed edges

A module declares its states, its parameters and its formalism: ODE or jump. It also declares how it touches the rest of the cell. INFERCELL has seven kinds of coupling edge (Table 2). Each kind says what crosses the edge and on which clock. Mass and currency edges act inside the ODE solve. Catalytic edges and deferred costs cross every second, at the handshake between the two blocks. Rate-constant edges cross every minute, when the stochastic block rebuilds its rate constants from the metabolite pools. The set of kinds is closed. A module cannot invent a new kind of coupling without a change to the framework.

Before anything runs, a resolver reads all the edges and builds the coupling graph. It checks the composition for about twenty kinds of error. Examples are a state that two modules integrate, an edge that names a species no module owns, and a moiety that is consumed but never produced. In complete mode it refuses to build a model with an unowned state or a stranded moiety. A composition that runs is therefore a composition that the framework has checked.

Table 2: **The seven edge kinds and their clocks.**

Edge	What crosses	Clock
Mass	a shared continuous state	ODE solve
Currency	flux through a shared pool (ATP, GTP)	ODE solve
Catalytic	a protein count into a rate law	1 s
Deferred cost	jump-side cost, debited one step later	1 s
Rate constant	pools into stochastic rate constants	60 s
Volume	membrane proteins into cell geometry, and back	each step
Clamped	a dependence replaced by a constant	declared only

2.3 Every reduction is declared

Any model of a cell leaves things out. In INFERCELL each departure from the published model is an object, not a comment. A reduction label has a category (one of 16, such as a clamp, a lumped step, a smoothed counter or an asserted prior) and a reason. The framework collects labels from three places: the edges, the parameter records and the modules. One call returns the full list for a composed model. We report this list in Section 3.8, and we measure the effect of only one entry on it.

2.4 One interface, two descriptions

Gene expression in INFERCELL has one interface and two descriptions.

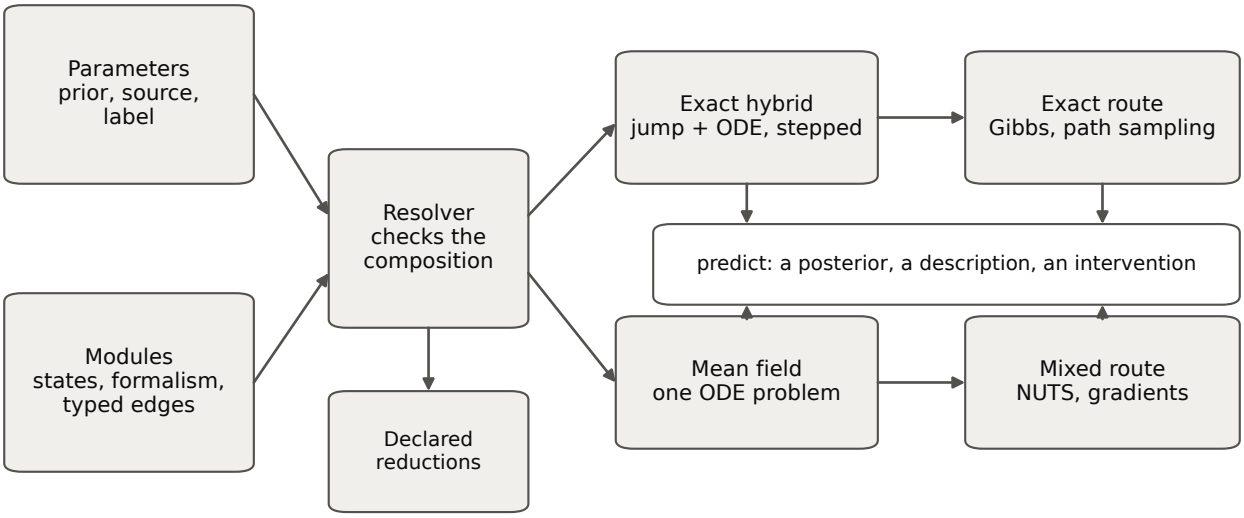


Figure 1: **The INFERCELL architecture.** Parameters carry a prior, a source and a label. Modules declare their states, formalism and typed edges. The resolver checks the composition and collects the declared reductions. The composed model runs as the exact hybrid or as one mean-field ODE problem, and each description has an inference route. `predict` takes a posterior from either route, a forward description and an optional intervention.

The exact hybrid. The stochastic modules run as a jump process, and the metabolic modules run as ODEs. A driver steps the two blocks in turn, as the published model does. Every second it writes protein counts into the metabolic rate laws, steps the ODEs, debits the costs of the stochastic events, and steps the jump process. Every minute it rebuilds the stochastic rate constants from the pools. We deliberately did not wrap this in one solver object. A single jump problem over an ODE problem would let the propensities follow the pools continuously. That is a different model from the published one. The driver can also record a stochastic path and replay it exactly. A replay matches the original run bit for bit at every handshake. The exact sampler (Section A.3) depends on this.

The mean-field swap. The gene-expression network of the model is first order. So, given the rate constants, the equations for the means of the transcript and protein counts close exactly, as linear ODEs. INFERCELL derives these equations from the same propensities and update rules that the jump modules use, so the two descriptions cannot drift apart. With the swap, the composed model becomes one ODE problem with callbacks for the one-minute rebuild. Forward-mode automatic differentiation [15] passes through this problem and its callbacks. We test the gradients against central differences to a relative error of 10^{-6}.

What is and is not differentiable. The exact hybrid is not differentiable, because the published model is not. The published model clips a drained pool at zero and rounds pools to whole molecules. Both make the model a step function of its parameters. The exact sampler therefore uses no gradients. The mean-field description gives gradients for the parts of inference that need them. The cost of the swap is a declared reduction, and we measure it (Section 3.3).

2.5 Inference across a boundary with no shared parameter

Three problems are often treated as one.

- *Local inference:* what do a module's own data say about its own parameters?
- *Uncertainty propagation:* given the uncertain output of one module, how uncertain is the next module?
- *Cross-module inference:* do the data of one module inform the parameters of another, when the two share no parameter?

The third problem is the hard one. Write u for the trajectory at the interface: enzyme counts and drains in one direction, pools in the other. For a pair of modules $A \to B$ with data D_A and D_B, the forward message is $p(u \mid D_A)$. The backward message is a likelihood over the interface. For a one-way coupling, with D_A and D_B independent given u,

$$\lambda_B(u) = \int p(D_B \mid \theta_B, u)\, p(\theta_B)\, \mathrm{d}\theta_B. \qquad (1)$$

The joint posterior over u is proportional to $p(u \mid D_A)\, \lambda_B(u)$. With feedback, as here, the messages alternate rather than close in one pass. A *statistical cut* drops λ_B [8]. It changes the target of inference, and data from

B can no longer inform A. A *dynamical reduction* instead removes a weak edge from the simulator and keeps full Bayesian inference. For cross-module inference, only the second kind keeps the information we want: metabolite data that inform a gene-expression parameter.

The standard modular methods pass messages through shared parameters: cuts and semi-modular inference [8, 10], and Markov melding [16]. In the reduced model of Section 3.1 no parameter is shared, so these methods have nothing to pass while we condition on the stochastic path. The exact sampler does what is left: it conditions on the path, and it updates the path with all the data. In this sense the exact sampler is modular inference with nothing dropped. When the paths are integrated out, a gene-expression parameter appears in both the transcript factor and the metabolite factor. The mixed route uses this structure.

2.6 Inference attaches to each data stream

A data stream declares its observation model. In this paper there are two streams: exact transcript counts per cell, and bulk metabolite measurements of the mean over many cells. INFERCELL has two inference routes.

- **The exact route.** A three-block Gibbs sampler conditions on the stochastic path. It updates the paths by conditional sequential Monte Carlo [17], the metabolic constants by slice sampling, and the gene-expression constants from exact one-dimensional conditionals (Section A.3).
- **The mixed route.** The paths are integrated out. The transcripts keep a birth–death likelihood given the mean-field pools, and the bulk metabolites get a Gaussian likelihood centred on the mean-field model, in the spirit of a synthetic likelihood [18, 19]. The route uses the gradients of the swap with the No-U-Turn Sampler [20]. It is an approximation, and Section 3.2 compares it with the exact route on M0.

One function, `infer`, runs inference, and `inference_route` chooses the route from the composition and the data. An all-ODE composition with time-series data takes the No-U-Turn Sampler, and an all-jump composition takes approximate Bayesian computation. A hybrid composition with cell and bulk data has the two routes above, and the user must name one, because one is a reference and the other an approximation. Anything else is refused, with an error that says why. The cell-data routes are built for M0's gene expression (Section 3.1), and we have demonstrated them only on M0.

2.7 From posterior to prediction

The function `predict` takes a posterior, a forward description and an optional intervention, and it returns a predictive distribution for a chosen quantity. The composition it predicts on may have parameters the fit never reached. We call those the held parameters θ_H, and the fitted ones the inferred parameters θ_I. The held parameters get only their priors and a viability rule V. The distribution is a cut in the sense of Plummer [8], with the first module a different model M_{fit} from the second,

$$p_{\text{cut}}(\theta_I, \theta_H \mid D, V) = p(\theta_I \mid D, M_{\text{fit}}) \times p(\theta_H \mid \theta_I, V), \tag{2}$$

and not a joint posterior under the second model. The data do not update θ_H, and θ_H does not feed back into θ_I. A variance split then says how much of the prediction's parameter uncertainty belongs to each block (Section A.7).

Which parameters are held is the user's choice. Freeing a fixed constant in its module's constructor adds it to θ_H, and `predict` draws it from its prior. Here a user releases lactate dehydrogenase's forward constant and records GTP's time course:

```
ms = d11_models()
ms[1] = CentralGlycolysis(
  enzymes = :translated,
  free = [:kcatF_R_ENO, :kcatR_R_ENO,
          :kcatF_R_FBA, :kcatR_R_FBA,
          :kcatF_R_LDH_L, :kcatR_R_LDH_L])
p = predict(ms, posterior;
  horizon = COREA_CYCLE_S,
  completion = :prior, n_held = 20,
  n_cells = 4, trajectory = [:M_gtp_c])
```

The first four names are the constants the reduced model already frees, and `n_held` is the number of draws of θ_H for each draw of θ_I. Reverse constants are derived from equilibrium constants and not drawn. Section 3.7 releases 62 constants this way. A release is a sensitivity to priors and not a posterior, since the fit held those constants at their nominal values.

3 Results

We follow one demonstration in three steps: a test system, the posteriors INFERCELL gives on it, and the probabilistic predictions those posteriors support. All results use synthetic data, and the priors are asserted. Results marked *unscored* are not compared with any truth. For the scored results, the test is calibration: over many datasets, a 90% credible interval must contain the truth about 90% of the time [21].

3.1 The test system: a reduced model of JCVI-syn3A, and M0

We test INFERCELL on a model reduced from the well-stirred JCVI-syn3A model of Thornburg and coworkers [3]. The reduction keeps the best-measured region of the cell and removes about 92% of it (Fig. 2).

- **Metabolism:** 22 reactions in four ODE modules: glycolysis to lactate, sugar transport and lactate

export, nucleotide recycling, and a lumped step for tRNA charging.

- **Gene expression:** 52 reactions in three jump modules: transcription, transcript decay and translation for 17 genes, and membrane insertion for the one membrane protein, the glucose transporter PtsG.
- **State:** 32 dynamic species and 5 held species, each named once in a registry with its initial value, prior width and source.

The two blocks couple in both directions, as in the published model. Enzyme counts set metabolic rates, and the costs of transcription and translation drain the nucleotide and charged-tRNA pools. In the other direction, the pools set the transcription and translation rate constants at each one-minute rebuild. Membrane protein counts set the cell's area and volume. No rate constant appears in both blocks. This property comes from the published model, not from our reduction.

The model makes no claim about JCVI-syn3A itself. With most of the cell removed, it cannot reproduce the doubling time, the metabolite levels or the proteome of the real organism. The code refuses to report a doubling time, so that no result can be compared with these quantities by mistake. The model's purpose is to give a real ODE–stochastic boundary from a published whole-cell model, at a size where an exact posterior is affordable.

An exact posterior on the full reduced model costs too much (Table 3). We therefore also use M0, the reduced model cut to two genes over 600 s. The two genes are GAPD, whose promoter is the best determined, and ptsG, the membrane transporter. All metabolism stays. The other enzyme counts are held at their measured copy numbers, and this hold is a declared reduction. M0 has six free parameters: the two promoter strengths, the transcript decay rate, the forward catalytic constants of enolase (ENO) and aldolase (FBA), and the bulk assay noise σ_{b}. All the posteriors in this paper are on M0.

3.2 Posteriors on M0

The posteriors. Figure 3A shows both routes on one M0 dataset of 50 cells. The exact reference is 4 chains of 1,650 sweeps. Its rank-normalised $\widehat{R}$ [22] is at most 1.007, and its bulk effective sample size is at least 1,081 for every parameter. The mixed route, which depends on a bulk covariance S computed once from a hybrid ensemble (Section A.5), uses 2 chains of 200 adaptation and 200 sampling steps per dataset. On this dataset the ptsG promoter, the decay rate and enolase sit 0.42–0.56 standard deviations from the exact reference, about 0.3 above the matched noise floor, the largest distance between chains run under identical settings. Re-estimating S at the posterior mean does not close this gap, and its cause is open.

Are they calibrated? A test against one known truth, as in DREAM8 [6], cannot show whether a posterior's width is right. So we drew 50 parameter sets from the prior, simulated one dataset from each, and fitted each dataset by each route, with one chain for the exact route. For each parameter we compared the empirical CDF of the ranks of the truth with a simultaneous band [23], at a joint level of 95% over the six parameters. On both routes all six parameters stay inside their bands (Fig. 3B), a test that detects only large miscalibration at this size. A sampler that returned the prior would also pass, so we note that the exact reference does contract: on its dataset the posterior variance is about 89–99.6% below the prior's. The largest deviation is 0.164 on the exact route and 0.183 on the mixed route, both for the transcript decay rate, where the band allows 0.244. A calibration test over many datasets does not exclude a gap of the size above on one dataset, so we report both. Section B.1 gives the coverage read from the exact ranks.

Table 3: **CPU-hours per posterior.** The mixed route's value is a median over completed jobs.

Route	M0	Reduced model
Exact, reference	6,810	—
Exact, production	507.5–678	∼21,000 (projected)
Mixed	6.57	not run

What they cost. A mixed posterior costs a median of 6.57 CPU-hours (range 5.04–13.08), against 507.5–678 for the exact sampler (Table 3). That median counts completed jobs only: a fault in the starting optimiser lost another 274.1 CPU-hours across the whole run. The long reference run cost 6,810 CPU-hours. A production-length exact chain, about one-13th of that, is not distinguishable from the reference on any marginal (Section B.1). An exact posterior on the full reduced model is projected at about 21,000 CPU-hours, above our bound of 1,000 CPU-hours per posterior. Exact inference therefore stops at M0. The mixed route costs far less on M0, but we have not calibrated it on the reduced model. The cost is the number of simulations, not the time of one. One cycle of the full reduced model takes 32.2 s, and a fast implementation of the full JCVI-syn3A model is the subject of a companion paper [24].

How closely the routes agree. Over the 50 datasets, the largest mean bias of the mixed route against the exact posteriors is 0.095 ± 0.050 posterior standard deviations, for the decay rate. The median ratio of standard deviations per parameter is between 0.92 and 1.07, and per parameter 9–22 of the datasets differ from the exact fit by more than three of its Monte Carlo standard errors.

Limits. The calibration results hold for M0 only. In 23 of the 50 exact replicates the smallest bulk effective sample size is below 100, and autocorrelated draws can distort ranks. In 17 of the 50 mixed replicates the two chains fall short of $\widehat{R} \leq 1.02$ and a bulk effective sample size of 100. The worst $\widehat{R}$ is 1.046 and the smallest effective sample size

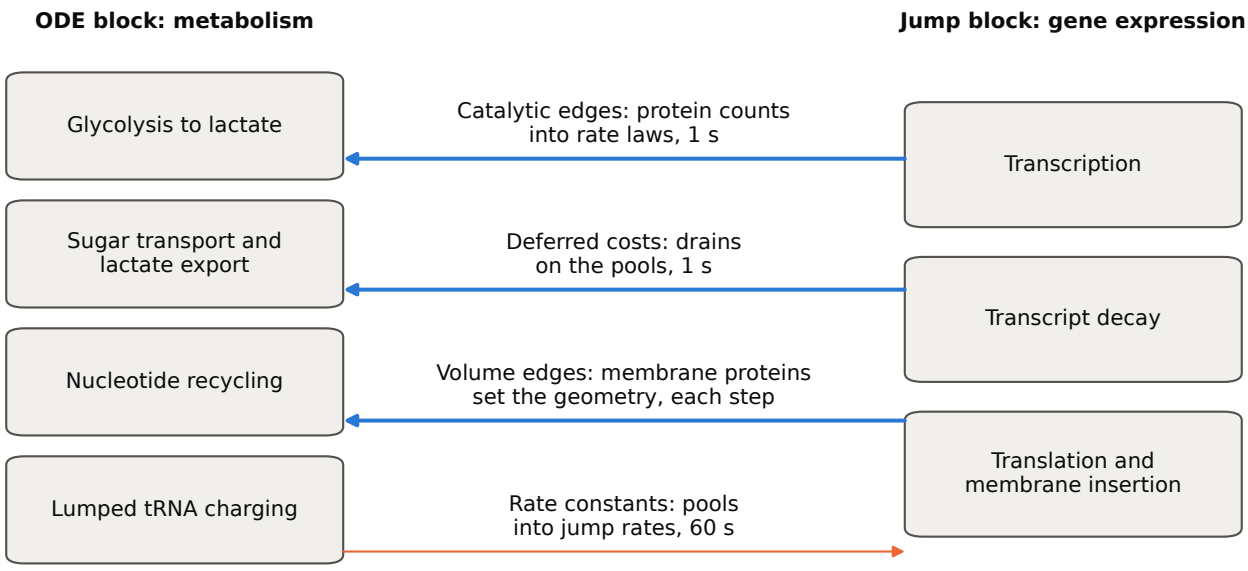


Figure 2: **The reduced JCVI-syn3A model.** The four metabolic modules (ODE) and three gene-expression modules (jump), and what crosses between them. Arrows run between the blocks, not the boxes they touch. Blue arrows run from gene expression to metabolism, and the drains have a gain of order one. The orange arrow is the weak return. Supplementary File S1 lists each departure from the published model.

is 81. The mixed calibration result is unchanged without those replicates, as a sensitivity check only, since dropping them is a selection.

3.3 What the mean-field swap costs

The mixed route rests on the mean-field swap, which replaces the stochastic gene expression with its mean. The means of transcripts and proteins are exact, but metabolism is nonlinear. A metabolism fed the mean drain differs from the mean over cells of metabolisms fed their own drains. The difference is a Jensen gap.

The gap is largest for GTP, where the mean field sits 10–16% below the population mean. We traced it to cell-to-cell variation in the drains. When every cell receives the mean field's drains, the gap closes. At our three test parameter sets, the enzyme counts, the per-cell feedback and the clip at zero did not account for the GTP gap. A possible mechanism, which we have not tested, is that only the saturable enzymes PGK3 and PYK3 regenerate GTP. If GTP is convex in the spend, the population mean sits above GTP at the mean spend.

We treat the gap as a declared reduction and correct it with an offset δ, the log difference between the mean of a reference ensemble and the mean field at one reference parameter set. Our reference is the most detailed composition we have, the exact hybrid, and not the truth about the cell. Without δ, the gap would move the transcript decay rate by 2.4–3.1 posterior standard deviations. With δ, the remaining bias has a root mean square within about one standard deviation, and single targets reach 1.9. Both figures are linearised. We report the gap with a goodness check (Section B.3) and with this bias, and we do not use it as a stopping rule. With δ in place, what remains at the level of the posterior is small on average over datasets, and larger on one dataset for a reason we have not found (Section 3.2).

3.4 What each data stream informs

Our claims rest on calibration on M0, and not on a measured transfer of information across the boundary by the exact sampler. Still, the posteriors say something about the boundary. A coupled model has three graphs, and they need not agree (Fig. 4). The *coupling graph* says which module drives which state. The *likelihood graph* says which likelihood factors depend on which parameters once the stochastic paths are integrated out. The *information graph* says which data streams move which parameters. In the reduced JCVI-syn3A model the coupling graph is two-way, but the information graph is sparse and points mostly in one direction.

The strength of each coupling direction predicts where information flows (Table 4). From gene expression to the pools the gain is of order one. The pools turn over in seconds to minutes, and the drains from gene expression are a large part of that turnover. From the pools back to the rate constants the gain is small: an elasticity of 0.044–0.051 for transcription and 0.091 for translation. The gain is small because the published rate laws are saturated. The elasticity here is $\partial \ln k / \partial \ln c$, of a rate constant k in a pool c. The published law's dissociation constant K_d for RNA

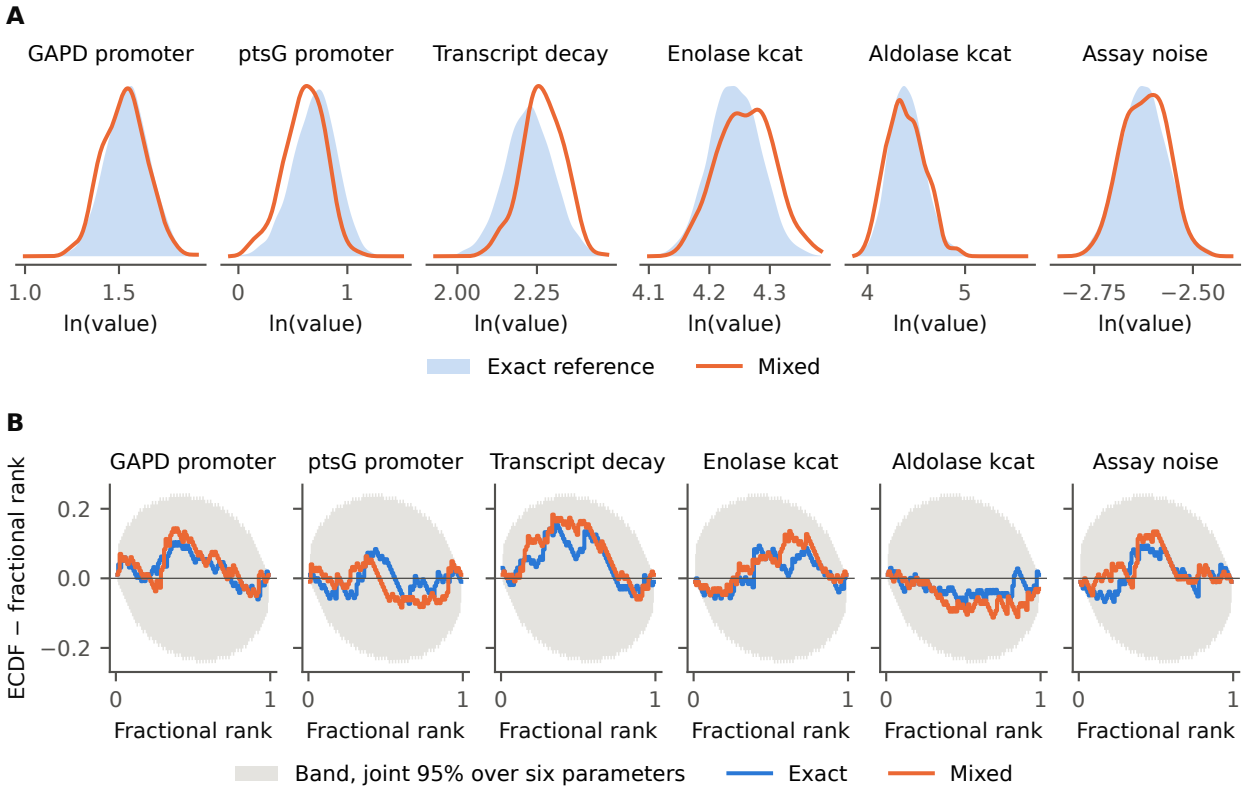


Figure 3: **Posteriors on M0, by both routes.** A: marginal posteriors of the natural log of each parameter on the reference dataset: the exact reference as a filled blue density, and the mixed route as the orange line. The ptsG promoter, the decay rate and enolase sit 0.42–0.56 standard deviations apart, and the cause is open. B: rank-ECDFs of both routes over the same 50 prior-drawn datasets, against simultaneous bands at a joint level of 95%. The grey region is the band, and each line is the rank-ECDF minus the fractional rank.

Table 4: **Gain and information in each direction across the boundary.** Information is the ratio of posterior to prior standard deviation at 200 cells. The ptsG value is a linearised bound at 10% assay noise, and its bracket is the ratio that noise alone gives. The ENO value limits what transcripts can inform, since they are a function of the path: a Gaussian approximation from the path's Fisher information, linearised at one truth, from 40 cells scaled to 200, with no assay noise.

Direction	Gain	Information
Gene expression → pools	order one	ptsG from metabolites only: 0.10 (bracket 0.19)
Pools → rate constants	0.044–0.051 (transcription), 0.091 (translation)	ENO from transcripts only: 0.999

polymerase and its nucleotides is 0.1 mM, while ATP is near 3.65 mM.

Metabolites to gene expression. The linearised bound says that metabolite data can inform the ptsG promoter, a gene-expression parameter that they never measure. On M0 the mixed route shows this shrinkage, to 0.676 of the prior standard deviation. That is less shrinkage than the bound of 0.47 projects, and that calculation's own split-half noise ratio is 0.68. The exact fit under metabolites only did not mix for this parameter, so on the exact sampler this direction is unmeasured, and we do not claim it there.

Transcripts to metabolism. Transcript data carry almost no information about the enolase constant, a metabolic parameter. We report that result as a null, not as a failure. On M0 at 50 cells, the exact sampler's divergence under transcripts only is 0.0041 nat for enolase and 0.0081 nat for aldolase, each below its noise floor. That read-off is M0's. We did not run the planned test at 1,000 cells, so we do not extend it to the full reduced model.

One data stream at a time, by both routes. On the reference dataset under transcripts only, none of the five marginals we compared differs detectably between the routes: the Wasserstein distance is 0.051–0.097 exact standard deviations, each below its noise floor. The decay rate's

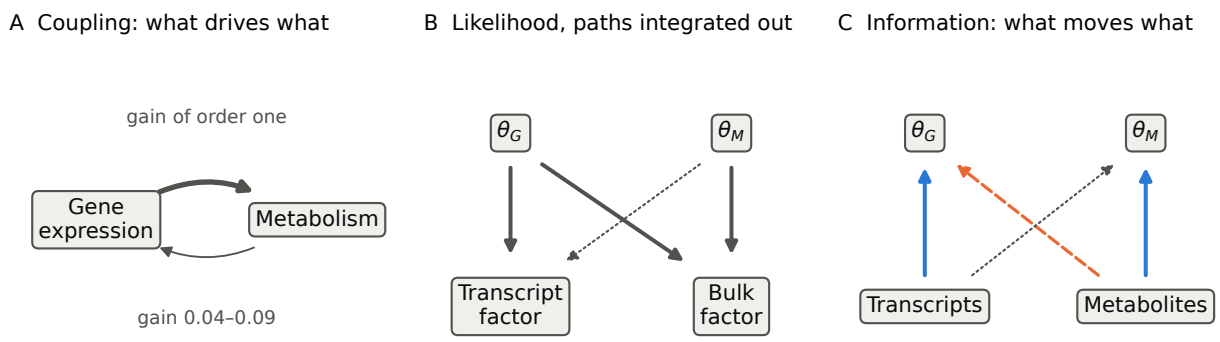


Figure 4: **Three graphs of one model.** A: the coupling is two-way, with a strong drive from gene expression to the pools and a weak one back. Line width shows gain. B: once the paths are integrated out, the gene-expression parameters θ_G enter both factors, and the metabolic parameters θ_M reach the transcript factor only weakly (dotted). C: what the data move. Solid: each stream informs its own module, on both routes. Dashed: metabolites to gene expression, shown by the linearised bound and the mixed route, with no exact result. Dotted: transcripts to metabolism, a null read-off on M0.

posterior variance shrinks by 98.9% under transcripts only: the autocorrelation in a time-resolved single-cell record separates it from the promoter strength, where RNA-seq data inform only the product of synthesis rate and half-life [6]. Under metabolites only, enolase, the decay rate and σ_b differ detectably, by small margins, and aldolase does not. The exact fit has an effective sample size of only 39 for the decay rate. Each floor is one sample of the noise, so these margins are not precise, and we have not tested whether they share a cause with the gap of Section 3.2.

3.5 Probabilistic predictions on M0, scored

What we predict. The quantity Q is the lactate a cell exports, in particles, and the estimand is its population mean $\mu(\theta)$. The intervention is a knockdown that scales the ptsG promoter strength by $1 - d$, at dose $d = 0.3$, with paired seeds, and its effect is ΔQ. We fixed the quantity, the estimand, the viability rule and the intervention on held-out pilots before any evidence run (Section A.7).

Why M0, and why scoring stops there. M0 is a slice of the reduced model small enough for the exact route to check a truth. It keeps all the metabolic reactions and the same blocks and code, with 2 of the 17 genes stochastic and inferred, and 12 enzyme counts and 3 carrier totals fixed, over 600 s. So this scores the posterior-to-prediction step, and not the reduced model's completion, its viability rule or its horizon. Cells also clip far less often here, in 0.77% of baseline cells. One exact posterior on the reduced model is projected at about 21,000 CPU-hours, and calibration needs 50 of them, about 1.0 million CPU-hours. We have not costed the same test by the mixed route.

The design. For each of 50 datasets with known truth, we pushed two posteriors through three forward models. The posteriors are the exact sampler's and the mixed route's (Section 3.2). The forward models are the hybrid, the mean field, and the mean field with an offset δ_Q for this quantity, taken from a new hybrid ensemble. On M0 nothing is completed from a prior, so there is no viability filter. We fixed the scoring before the run. With 100 draws, a calibrated arm's intervals cover 88.2% at nominal 90% and about 49% at nominal 50%, with standard errors over 50 datasets of about 4.6 and 7.1 points. Those are the references we compare with. They are approximate: thinning lowers them slightly, and the hybrid's Monte Carlo term raises its own coverage, to about 92% at nominal 90%. Figure 5 shows the predictions themselves with the hybrid, and Fig. 6 and Table 8 score all six arms.

Lactate export. The exact posterior with the hybrid, and the mixed posterior with the hybrid, cover at the reference (Fig. 6). The exact arm is high at nominal 50%, by 1.8 standard errors. The mixed route's intervals are about 18% narrower. Changing the forward description shifts the prediction by 0.14 standard deviations at the same width, and δ_Q removes that shift. Changing the inference route shifts it by -0.11 standard deviations. The mixed route's narrower intervals sit beside marginal standard deviations close to the exact ones (Section 3.2), and we have not examined whether the routes differ in the dependence between parameters. The cheapest route, the mixed posterior with the mean field, covers 82–84% at nominal 90%, about one standard error low, with a z-score standard deviation of 1.17–1.21. Its small shift without δ_Q is the two effects nearly cancelling, not the absence of either. 50 datasets cannot separate these arms at 90%: all six are within 1.4 of the reference's standard errors. The mixed posterior's sizes sit on the limit, with the sizing rules below missed on 25–27 datasets.

The knockdown effect. A model fitted to unperturbed data predicts the effect of the knockdown with the right

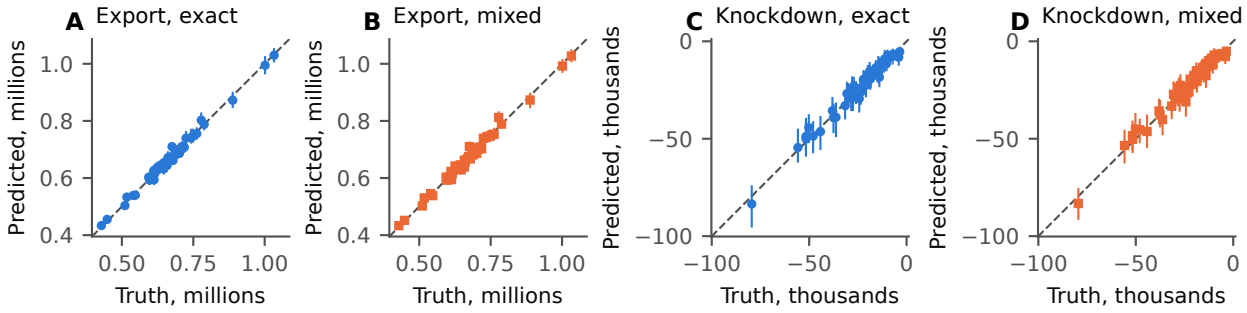


Figure 5: **Probabilistic predictions on M0, against the truth.** Each point is one of 50 datasets, with the predictive mean and 90% interval from the hybrid, and the dashed line is equality. A, B: lactate export. C, D: the knockdown's effect, whose coverage does not meet the sizing rules in the text. Blue circles use the exact posterior and orange squares the mixed. The truths were drawn from the prior, so they span a range wider than the intervals, and closeness to the line reflects that range too. Coverage (Fig. 6) is the test.

sign, and z-scores with standard deviations of 0.74–1.27. The sign is a weak test, because less transporter is expected to lower export. The true effect has a median of $-20{,}400$ particles, and every arm's 90% interval excludes zero with the right sign on all 50 datasets. We read this off after the run. These coverages are not a clean calibration. We sized the run on Q, with the rules that cell noise be at most 10% of the spread and the truth's standard error at most 10% of the predictive standard deviation. The panel fails both rules for ΔQ on all 50 datasets, because pairing removes little of the cells' spread: the noise share has a median of 0.503 under the exact posterior. By arithmetic, and not by a measurement, that noise accounts for the hybrid arms being too wide. The mean-field arms carry no such term, so the truth's error counts against them. The mixed posterior with the mean field is the one arm well below the reference: $32.0 \pm 6.6\%$ and $26.0 \pm 6.2\%$ at nominal 50%, and $78.0 \pm 5.9\%$ at nominal 90% with δ. Noise explains part of that, and the rest cannot be separated from that route's narrower intervals at these sizes. Meeting the rules would cost on the order of 100,000 CPU-hours, and we did not run it.

What this supports. On M0, the posterior-to-prediction step covers at the expected rate on Q, and it gets the sign of the knockdown right. Q is not an unmeasured quantity. The data include extracellular lactate, a noisy bulk measure of the export, so Q's coverage tests a prediction of something the fit saw, and we have not measured how far that stream drives the posterior. The knockdown's effect ΔQ is what the data did not see. This panel is the only validation behind the reduced model's knockdown prediction below. It does not test the transfer from M0 into the reduced model.

3.6 Probabilistic predictions on the reduced model, unscored

We now predict the same quantities on the reduced model, over one published cycle of 6,300 s, against 600 s on M0. No truth is checked here.

The parameters are a modular transfer. We use the cut of Section 2.7 with M0 as the fitted model. The five inferred parameters θ_I come from one exact posterior on M0, one of the fits of Section 3.2, with the assay noise left out. Fifteen promoter strengths θ_H belong to genes that M0 removes. We use the cut because an exact posterior on the reduced model is out of reach (Table 3). M0 fits θ_I with the counterparts of θ_H fixed, so this transfer assumes that the posterior of θ_I would not move if they varied. We have not tested that assumption.

The prediction and its budget. We ran 100 draws of θ_I crossed with 20 draws of θ_H, at each of three prior widths (Table 5 and Fig. 7). At the default width, among the 20 parameters drawn and on this one posterior's θ_I, about $91\% \pm 4\%$ of the parameter uncertainty in lactate export comes from the 15 promoter strengths that M0's fit cannot reach. Figure 7B shows this directly: the spread across draws of θ_H grows with the prior width, and the spread across draws of θ_I changes much less. Their medians are a proteomics proxy, and their spread (geometric standard deviation 2) is our choice. The share runs from 63% at half that log-scale width to 98% at double. The two nested orders of the split differ by about a point or less. Every other constant is held fixed or derived, so this is a split among these 20 parameters only, and it is not a property of the model alone.

The share is of the variance of μ. A new cell's variance also holds the cell's own spread, and of that total the held parameters' part at the default width is 72%. The predic-

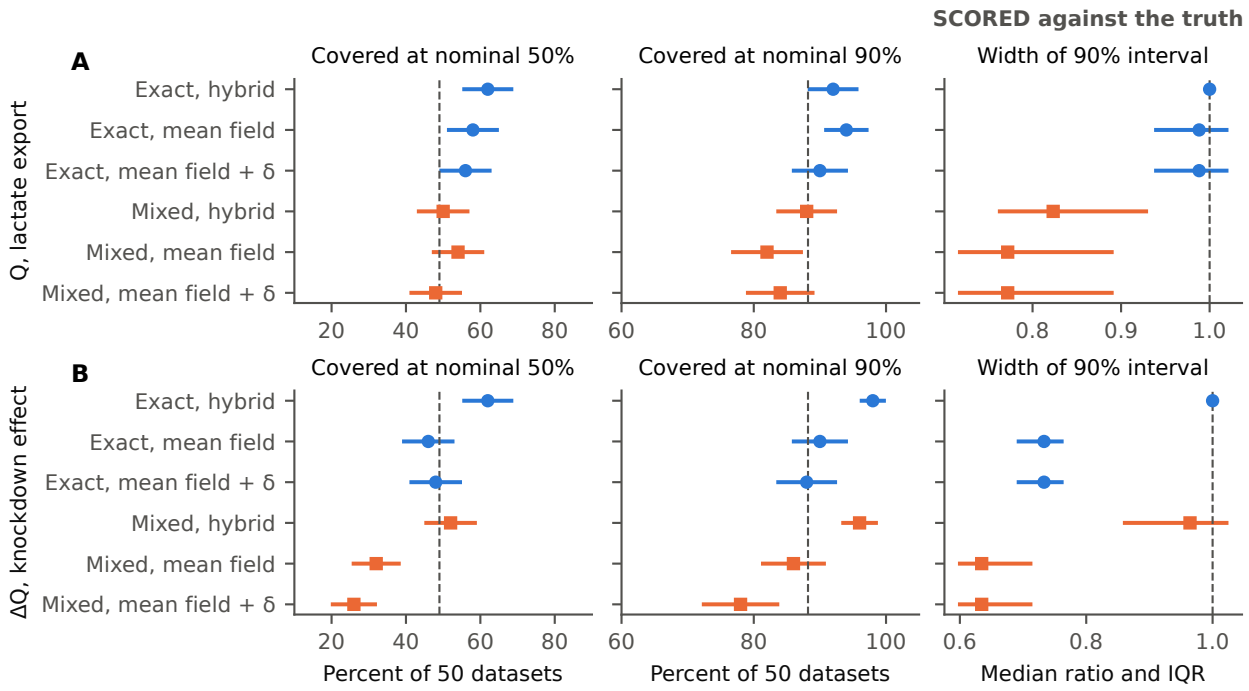


Figure 6: **The posterior-to-prediction step scored on M0.** Each arm is a posterior and a forward model, over 50 datasets, with 100 draws per posterior. Blue circles are the exact posterior and orange squares the mixed. Coverage is ± one standard error, and the dashed lines are the references of 49% and 88.2%. A: lactate export. B: the knockdown's effect, whose coverage does not meet the sizing rules in the text. Width is the median ratio to the exact posterior with the hybrid, with its interquartile range.

tion itself moves with the prior too: mean export rises by a factor of 1.459 ± 0.079 from the narrowest width to the widest.

The knockdown. At the default width, mean lactate export falls by $1.03\% \pm 0.14\%$. This effect shrinks as the prior widens, and at the widest setting its interval of two standard errors reaches zero. At 4 cells per pair, the run cannot resolve the parameter variance of ΔQ, because cell noise swamps it. So we report its mean and no variance split for it.

Viability and validity. The viability rule rejected nothing at the two narrower widths, so there the conditioning removed nothing. At the widest it removed 6.45% of pairs, all from three of the 20 draws of θ_H. Clipping is the common state in this run, and the rule allows it: most cells clip a pool at least once (Table 5). A likely cause is this posterior's transcript decay rate, which is low against the reference dataset's, though not an outlier among the 50 fits. Across the decay rate's deciles at the default width, the clipped fraction falls from 84% to 61%. We have not shown this link on stochastic cells. Nearly every cell's GTP falls at least once below K_d, the regime where the small elasticities of Section 3.4 are in question. The run stores no elasticities, so it shows the pool reaching that regime and not the basis failing. These figures, and the parameter variance, hold for this one posterior, and another could give different ones.

What this does not show. It is not calibration, on the reduced model or anywhere else, and it does not compare the reduced model with real measurements. The posterior of θ_I is a fit to M0's synthetic data over 600 s, so its small share is not what real measurements would leave.

3.7 Predictions over time, and releasing a constant

Figure 8 follows the same predictions through time. The runs replay the designs of Sections 3.5 and 3.6 unchanged, and `predict` records chosen species every 60 s in the cells it already simulates, so the bands need no new inference.

On M0. Row A is one dataset, replicate 1 with the exact posterior, chosen because this posterior feeds the reduced model. The truth lies inside each 90% posterior band at every recorded time after the start. One dataset is an illustration, and the test remains the coverage over 50 datasets. Lactate export is not an out-of-sample quantity. The data measure extracellular lactate, whose only dynamics are the export, so the figure overlays those data on export's axis. Pyruvate is not in the measured panel. Its band is at most 59 particles wide at a level of about 91,000, under 0.07%.

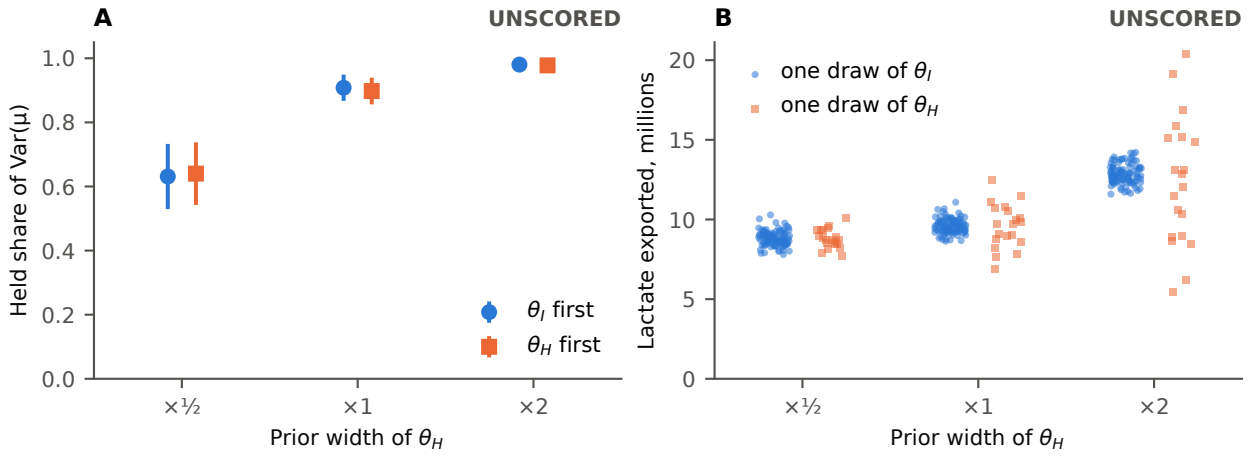


Figure 7: **What the reduced model's prediction rests on (unscored).** No truth is compared with anything here, and all of it rests on one exact M0 posterior's θ_I, with 100 by 20 parameter pairs over 6,300 s. A: the share of $\mathrm{Var}(\mu)$ held by the 15 promoter strengths, in both nested orders, $\pm$ one standard error. B: predicted lactate export per parameter draw. A blue circle is one draw of θ_I, averaged over the draws of θ_H, and an orange square is one draw of θ_H, averaged over the draws of θ_I. Pairs the viability rule rejected are left out, and horizontal jitter is for legibility. Each pair has 4 cells, so the points also carry some cell noise.

Table 5: **The reduced model's uncertainty budget. Unscored**, on one exact M0 posterior's θ_I. Width scales the log-scale prior width of θ_H at a fixed median. Q is lactate exported over 6,300 s, in millions of particles, and ΔQ is the mean effect of the knockdown in thousands. Held share is the share of $\mathrm{Var}(\mu)$, with θ_I first in the nested split. Cell rows count the 8,000 baseline cells before viability filtering. A starved cell has also clipped.

Width	$\times\frac{1}{2}$	$\times 1$	$\times 2$
Geometric SD of θ_H	1.41	2	4
Mean Q	8.80	9.58	12.84
$\sqrt{\mathrm{Var}(\mu)}$	0.75	1.47	3.84
Held share	0.631 ± 0.102	0.908 ± 0.041	0.980 ± 0.009
Pairs kept by V	1.000	1.000	0.9355
Cells clipped	70%	74%	90%
Cells starved	0%	0.35%	8.8%
Mean ΔQ	-118 ± 9	-98 ± 14	-40 ± 32

On the reduced model. Row B splits the band into its layers. The draws of θ_I alone give the narrowest band. Early in the cycle the spread of a new cell widens it as much as θ_H does, or more, for GTP until about 20 minutes. By the end of the cycle θ_H adds most of the width, and a new cell adds little more. The middle layer also holds the Monte Carlo noise of 4 cells per pair, about 6% of its variance at the end of the cycle. In the first 6 minutes that noise exceeds the parameter variance, so we do not show the variance split there. After that, the share of $\mathrm{Var}(\mu)$ from θ_H stays between 0.84 and 0.91. GTP's band over pairs reaches K_d in the second half of the cycle, and the band over single cells from about 41 minutes. A band at each time is not the share of cells that are ever below K_d, which is most of them (Section 3.8).

Releasing a constant to its prior. A user can free any fixed constant, and `predict` then draws it from its prior with the other held parameters (Section 2.7). Row C releases all 62 fixed forward rate and Michaelis constants of the two metabolic modules, so 82 parameters are drawn. The viability rule kept 1,246 of 2,000 parameter pairs, 62%, and 38% of all cells starved. Without the rule, the lower edge of export's band flattens near 0.65 million particles within minutes, and the median prediction falls by 11%. With it, the median is within 3% of row B's, and the band of export at the end of the cycle widens from 4.9 to 7.1 million. GTP moves far more. Its mean at the end of the cycle is about 25,000 particles, against 4,800 in row B, and the upper edge of its band sits near 40,000 throughout. We have not traced what sets that edge. This is a sensitivity to the priors of constants that were never fitted, not a posterior over 82 parameters, and it stresses the cut harder than row B, because M0's posterior was fitted with these constants at their nominal values.

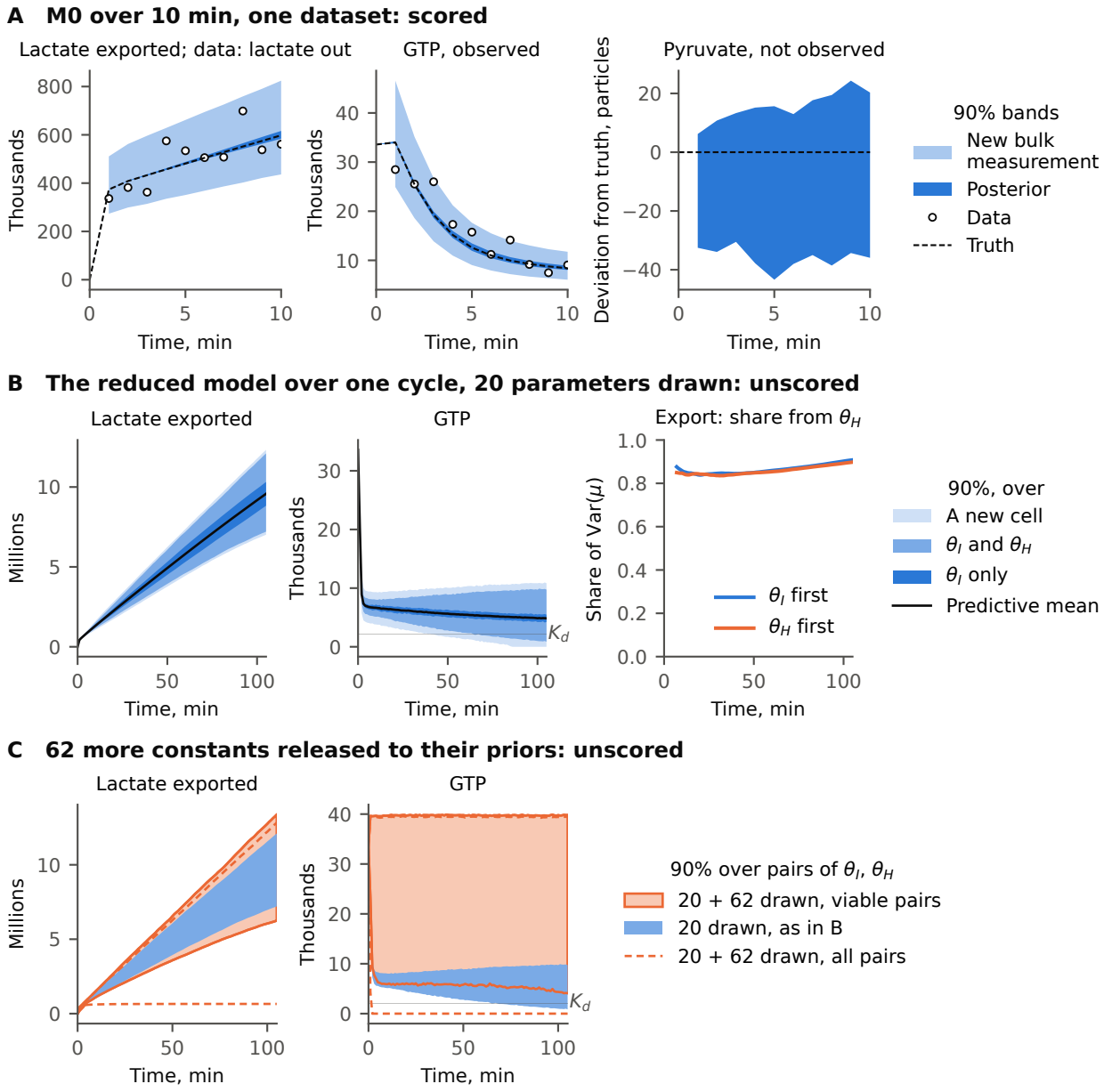


Figure 8: **Predictions over time.** All data are synthetic, and no band predicts a real cell. Bands are 90%. A, scored: M0 over 600 s on one dataset, with the exact posterior. Extracellular lactate and GTP are observed, and the data, regenerated by `m0_dataset(16_200_001; ncells = 50)`, are shown with a band for a new bulk measurement. Extracellular lactate is put on export's axis by one constant ratio, whose largest deviation was 0.25%. Pyruvate is not observed and is shown as its deviation from the truth. Calibration is the coverage over 50 datasets (Fig. 6), not this one. B and C, unscored: the reduced model over 6,300 s, with θ_I from the same posterior and 20 parameters drawn. Row A does not test the transfer into this model. B: the bands over draws of θ_I alone, over pairs of θ_I and θ_H, and over single cells, with the share of $\mathrm{Var}(\mu)$ from θ_H in both nested orders. The grey line is GTP's K_d. C: 62 more constants drawn from their priors, over pairs the viability rule kept (filled) and over all pairs (dashed), against row B's pairs. This is a sensitivity to priors, not a posterior.

3.8 What the predictions leave out: model form

A prediction's uncertainty has four sources: inferred parameters, held parameters, intrinsic noise and model form. The first three have numbers in Section 3.6. Model form does not have one. Each reduction is listed in Table 6 and marked by what we know of its effect on Q. Only the formalism swap is measured, because only it has a column

in the crossed design. Every other reduction is declared and not bounded, so the budget does not cover the full uncertainty a modeller should carry.

The one-way reduction on the reduced model. The elasticities that justify this reduction hold while GTP stays well above K_d. At the default width, 94.7–95.8% of baseline cells fall below it at least once, and 69% reach exactly zero. Clip counts give a lower limit on the time spent below it: 262 of 8,000 cells spend more than 21% of their post-debit instants at zero. This bound says nothing about time between zero and the threshold, so it undercounts. An earlier screen, with the inferred parameters drawn from their priors rather than this posterior, found GTP below the threshold at 21–55% of rebuild instants in 12 of 20 draws, and at none in 6. On those draws the basis does not hold over the full cycle, whatever the dose. On M0 the same screen found GTP below the threshold at no instant at the chosen dose. So this caveat applies to the reduced model and not to the scored system, and the scored check does not cover it.

4 Discussion

INFERCELL carried one demonstration from a test system through posteriors to predictions. On M0, both inference routes pass simulation-based calibration, with small declared discrepancies, and the posterior-to-prediction step covers lactate export at the expected rate, though the data measure export noisily through extracellular lactate, and how far that drives the posterior is unmeasured. A knockdown's effect, which the data did not see, has the right sign but not a clean calibration. On the reduced model the predictions are unscored, and their budget depends strongly on a prior width that we chose. Any user can release fixed constants to their priors, and releasing 62 of them raises the predicted mean GTP at the end of the cycle about fivefold among viable pairs, unscored and on one posterior (Section 3.7). The M0 check does not test the completion or the transfer into the reduced model.

What scales and what does not. Exact inference is calibrated on M0, and it stops there. The cost of an exact posterior grows with the number of genes and the length of the record, and the full reduced model is already above our budget. The mixed route removes the stochastic paths from the sampler. Two quantities limit it. The first is the Jensen gap, which we declare and correct (Section 3.3). The second is $\mathcal{R}$, the variance of the bulk log-likelihood over the stochastic paths. As an upper bound, at nominal parameters, $\mathcal{R}$ is about 64 on M0 at 50 cells and 70 on the full reduced model at 200 cells, both at 10% assay noise. If $\mathcal{R}$ is near those bounds, the metabolites constrain the paths together, and a single draw of paths per parameter value is not enough. On M0, the Gaussian bulk factor passes calibration in this regime (Section 3.2). We have not tested it on the full reduced model.

The observation model sets the interface. The three graphs of Section 3.4 suggest a working rule, from one model and two directions. Single-cell data inform the parameters of their own module. Information between modules travels on bulk observables, and it follows the gain of the coupling. A modeller can compute the gains before any inference. In this work they anticipated the transcript null. For the other direction the evidence is thinner: on one dataset the mixed route shrinks the ptsG promoter by less than the linearised bound projected, and the exact sampler did not measure it.

Reductions as approximations to inference. A dynamical reduction changes the simulator and keeps full Bayesian inference. A statistical cut keeps the simulator and changes the target. We prefer the first, because its cost can be measured against a more detailed composition. The formalism swap is such a reduction. We do not claim that the more detailed composition is the truth about the cell. We claim that each choice of description is declared, and that the swap's cost is measured on M0.

The prediction of Section 3.6 uses the other tool, a cut, and it does so on purpose. There the choice is not between two ways of inferring one model. No data reach the held parameters at all, and an exact posterior on the reduced model is out of reach. So the cut says plainly what the prediction is: a modular transfer of an M0 posterior, with the other parameters at their priors. Its cost is not measured, and it is declared (Table 6).

4.1 Limitations

- *Synthetic data only.* The data come from the model itself. Real assays need observation models for unknown cell numbers in bulk metabolomics and for capture efficiency in single-cell RNA counts.
- *One organism, one reduction.* We demonstrate the framework on reduced JCVI-syn3A only, and it has no generic organism registry. Generality across organisms needs a second implementation.
- *Asserted priors.* Many priors have no width in their source, so we set the width. This choice matters: the held share of the budget runs from 63% to 98% across the widths we tried (Section 3.6).
- *Exact calibration at small scale only.* Coverage of the exact sampler at the full reduced model is out of reach at about 21,000 CPU-hours per posterior. The reduced model's prediction is therefore unscored, and it rests on one posterior and on a transfer we have not tested.
- *Weak reverse coupling.* The mixed route relies on the small elasticities of the rebuild (0.044–0.091). On the reduced model about 95.8% of baseline cells already fall below K_d at least once (Section 3.8), so this basis is in question there without any perturbation.

Table 6: **The model-form table.** Counts come from `reduction_declarations` on the reduced model (93 departures) and on M0 (80). Supplementary File S1 lists each one, with two driver-policy records that are not departures. Effect is the effect on Q.

Reduction	Applies to	Effect
Formalism swap: stochastic gene expression replaced by its mean	Both	**Measured** on M0: a shift of 0.14 standard deviations at the same width, which δ removes (Section 3.5)
Transfer of the M0 posterior into the reduced model, where the proteins M0 holds are expressed stochastically	Reduced model	Declared, not tested. A test (fit M0 to reduced-model data by the mixed route, predict, and score) is neither designed nor costed
12 enzyme counts held at proteomics copy numbers, and 3 carrier totals at their initial values	M0	Declared, not bounded
One-way nucleotide reduction: the model keeps both directions, but the mixed route and our reading of information treat the pools' effect on rate constants as weak, on elasticities of 0.044–0.091	Both	Declared, not bounded, and in question on the reduced model: see the text
Clamped CTP and UTP pools (2)	Both	Declared, not bounded
Lumped tRNA charging, in the ODE block (1 lumping, 1 formalism)	Both	Declared, not bounded
Asserted priors (52 on the reduced model, 37 on M0), and priors the driver overwrites (16)	Both	The 15 promoter widths are varied in Table 5; the rest are declared, not bounded
Membrane growth: frozen baseline area and a calibrated footprint, with capped geometry (1, 1, 1)	Both	Declared, not bounded
Other module notes (18 on the reduced model, 20 on M0)	Both	Declared, not bounded

4.2 Future work

Four directions follow. The first is the method's own limits: conditioning the bulk factor on the observed transcripts, sampling paths only for pools near the one-particle floor, and handling strong-feedback regimes. The second is scale and data: the full well-stirred JCVI-syn3A model through the fast port of the companion paper [24], then real measurements. The third is comparing model structure and formalisms, which the swap makes possible, by predictive scores rather than Bayes factors. The fourth is goal-directed inference and experimental design: which parts of a coupled simulator must be kept, and which latent variables inferred, to keep the information a prediction needs?

Supplementary material

Supplementary File S1: every declared reduction. The output of `reduction_declarations` for the reduced model and for M0, by composition, category and subject, in `paper/tables/S1_File_reductions.tsv`.

Software and data availability

INFERCELL is open source at `https://github.com/MACSYS-CoE/InferCell`, with documentation at `https://macsys-coe.github.io/InferCell/`. The version used in this paper, and the evidence archive behind its predictions, are deposited with the v0.3.0 release at [Zenodo DOI, from the release]. The archive holds the prediction runs' outputs and the posterior fits they read, with SHA-256 checksums. The repository holds the rank tables and chain summaries behind the calibration figures, in `dev/scripts/` and `paper/figures/data/`, which also holds the bands of Fig. 8, copied from its run. Each number in this paper is listed with its source file in `paper/numbers.tex`, and each run's result note in `dev/scripts/` gives its commit and Slurm jobs. All data are synthetic.

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All computation ran on the OzSTAR cluster. [Thanks to collaborators, if any.]

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Appendix A Methods

A.1 The hybrid driver

The driver keeps two blocks with separate state: metabolite pools in mM as floating point values, and molecule counts as integers. One handshake step does the following, in order:

1. update growth and cell geometry from the membrane protein counts;
2. write the protein counts into the metabolic rate laws (catalytic edges);
3. step the ODE block by 1 s (Rodas5P, absolute tolerance 10^{-10}, relative tolerance 10^{-8});
4. debit the pools by the costs of the stochastic events of the previous step (deferred-cost edges), clipped at zero as in the published model;
5. every 60 s, rebuild the stochastic rate constants from the pools;
6. step the jump block by 1 s with Gillespie's direct method [25].

On the jump clock the rebuilt constants therefore take effect at $60m - 1$ s. Pools are written back as molecule counts under a declared rounding policy. A recorded path can be replayed: the replay applies the recorded firings instead of drawing new ones, and it returns the log density of the path and its sufficient statistics (event counts and exposures).

A.2 Observation model

Each synthetic dataset has C cells (50 on M0), observed every 60 s.

- **Transcripts:** the exact count of each gene's transcripts in each cell at each save.
- **Bulk metabolites:** one measurement per pool per save of the mean over the C cells, $\bar{x}$, with log-normal noise: $\log z \sim \mathcal{N}(\log \bar{x}, \sigma_{\mathrm{b}}^2)$. The assay noise σ_{b} is a free parameter.

We use bulk metabolites because a per-cell metabolite panel cannot be measured. The panel has 17 pools: G6P, F6P, intracellular and extracellular lactate, G3P, DHAP, charged tRNA, phosphate, 3PG, 2PG, FDP, ATP, ADP, AMP, GTP, GDP and GMP. The prior on the assay noise is $\sigma_{\mathrm{b}} \sim \mathrm{LogNormal}(\log 0.2,\ 1)$.

A.3 The exact sampler

The sampler is a three-block Gibbs scan with the stochastic path as a latent variable.

1. **Paths.** Conditional sequential Monte Carlo over 60 s windows [17], with particle Gibbs at $N = 20$ particles. Each particle is a snapshot of the driver. Transcripts are proposed as exact endpoint-conditioned birth–death bridges [26], so every proposal agrees with the observed transcript counts. The other reactions are then simulated given the transcripts. The weight is the bulk metabolite likelihood times each gene's transition probability. Cells are updated in turn, because the bulk likelihood couples them.
2. **Metabolic constants and σ_{b}.** A univariate slice step on each log parameter, given the paths. Reverse constants follow from the forward ones and the equilibrium constants.
3. **Gene-expression constants.** Given the path, each promoter strength and the decay rate have an exact one-dimensional conditional of the form $nu - e^u A + \log p(u)$, where n is the event count and A the exposure. We sample it by a slice step.

The sampler is written directly, not in a probabilistic programming language. The path update needs ancestor sampling and exact snapshots of the driver, and a general framework could not provide them without sharing solver state between particles. Runs can be checkpointed and resumed bit for bit, and the threaded sweep is bit for bit equal to the serial one.

A.4 The transcript factor

When the paths are integrated out, each gene's transcripts form a birth–death chain with a piecewise-constant birth rate, observed exactly every 60 s. The likelihood of one cell's record is a product of transition probabilities. We compute them by uniformisation with tridiagonal products, truncated where the remaining mass is below 10^{-12}. The cap on the count is fixed per dataset, and the number of terms comes from the primal value, so the factor is differentiable in the rates.

A.5 The mean-field route

The swap replaces the three jump modules with one mean-field module. Its equations are generated from the jump modules' own propensities and update rules. The composed model is one ODE problem. Four edges change form, and each change is a declared reduction:

- catalytic edges read the mean protein counts;
- drains are continuous and not clipped, and the solve stops if a pool becomes negative;
- the rebuild is a discrete callback at $60m - 1$ s and at $t = 0$;
- growth is continuous.

The offset δ plays the part of a model-discrepancy term [27], fixed at one reference value. The bulk factor is $\mathcal{N}\big(\log z;\ \log \max(\bar{x}_{\mathrm{MF}}(\theta), 1) + \delta,\ \sigma_{\mathrm{b}}^2 I + S\big)$, with δ and S from an ensemble of the exact hybrid at a reference parameter value, and with $\bar{x}$ floored at one molecule. The

route multiplies this factor by the transcript factor, so it ignores that both read the same cells. The reference value is the maximum a posteriori estimate at $\delta = 0$ and $S = 0$, recorded per dataset. The ensemble has 800 cells, on seeds disjoint from the dataset's. We take δ as the log difference between the ensemble's mean and the mean field at the reference value, per pool and save. We take S by the delta method, $S_{ij} = \mathrm{Cov}_{ij}/(C\bar{x}_i\bar{x}_j)$, for the covariance Cov of the per-cell panel counts. On M0 it is full rank, with 17 pools at ten saves.

The sampler is the No-U-Turn Sampler on the log of the six parameters, with the exact sampler's priors. Each fit has 2 chains of 200 adaptation and 200 sampling steps, started 0.25 prior standard deviations from the reference value. A parameter value outside the transcript factor's support, or a solve that does not succeed, has a log density of $-\infty$.

A.6 Simulation-based calibration

For each of $R = 50$ replicates we drew a parameter set from the prior, simulated a dataset, and ran one chain of 656 sweeps from a prior draw. We dropped 100 sweeps and thinned the rest to $L = 100$ draws. For near-independent draws, the rank of the truth among them is uniform if the sampler is calibrated [21]. We tested each parameter's rank-ECDF against a simultaneous band at $1 - 0.05/6$ [23], so the test as a whole is at the 95% level. The pass rule was fixed before any rank was computed.

A.7 Prediction

What was fixed in advance. We fixed the quantity Q, the estimand, the viability rule, the dose and the variance split on pilots with held-out parameters and seeds, before any evidence run, and recorded them in the project's amendment log. The pilot's sensitivity rule for Q asked θ_I to hold at least 5% of the parameter variance at 6,300 s, and lactate export did not meet it (3.3% $\pm$ 1.5%). We kept Q, on a reading written after a small trial run and before the full pilot: the failure counts against Q only if θ_I's elasticities are under a quarter of θ_H's, and they are not (0.056 against 0.047). That reading suggests the small share comes from the tight posterior.

Estimand. For a parameter draw, $\hat{\mu}(\theta)$ is the mean of Q over n hybrid cells with fresh seeds. Its noise is $\sigma^2(\theta)/n$, and we subtract the mean of that term from the spread over draws. The intrinsic spread is reported separately, from the hybrid only.

Completion and viability. On the reduced model, θ_H is drawn from its prior, and width scales its log-scale standard deviation at a fixed median. A cell is failed, starved, below a floor, clipped or clean, in that order of precedence, and a cell is starved if some counter clipped at more than half its drains. A cell is viable unless it failed, starved or fell below a floor, and we set no floors, so clipped cells count, and a parameter pair is viable if at least half of 4 separate cells are. We condition θ_H on viability per draw of θ_I, so θ_I's marginal stays the M0 posterior, given at least one viable θ_H for it. A joint variant, which also reweights θ_I, gave the same shares to the reported precision at the one width where the rule acted.

The variance split. The held share with θ_I first is

$$\frac{\mathrm{E}_{\theta_I}[\mathrm{Var}(\mu \mid \theta_I)]}{\mathrm{Var}(\mu)}, \tag{3}$$

and with θ_H first it is $\mathrm{Var}_{\theta_H}(\mathrm{E}[\mu \mid \theta_H])/\mathrm{Var}(\mu)$ [28]. The gap between them is the interaction plus any dependence the rule induces. With a crossed matrix of $\hat{\mu}$ we use weighted U-statistics on the centred values, and we report both orders. We checked the estimator's bias by simulation on known splits. The standard errors are a two-way jackknife, checked on synthetic matrices with the same structure. Near a share of one that check failed narrowly, so the widest setting's standard error may be too wide by up to about half.

The intervention. A knockdown at dose d scales the ptsG promoter strength by $1 - d$ after the parameter draw, and the effect is paired on common seeds. We assume that the intervention acts only through that parameter, that the posterior fitted to unperturbed data carries over unchanged, and that the model stays valid at the dose, which the screen below supports on M0 and which is in question on the reduced model (Section 3.8). V is read at baseline: a rejected pair is not run under the intervention, and a cell that fails only under it is counted as part of the prediction. The largest dose in $\{0.1, 0.2, 0.3\}$ that kept M0's pools in their regime was 0.3.

Scoring on M0. Each posterior contributes 100 draws at 887 cells each. The truth and the offset δ_Q, a difference in Q itself and not the log offset of Section A.5, each use 9,849 cells. Coverage counts the truth's estimated value inside the central interval of the draws.

A.8 Compute

All runs used the OzSTAR cluster, and the M0 scoring and the budget ran on its AMD `znver3` nodes. Table 7 lists the Slurm jobs behind the main results. The commits and checksums are in the evidence archive and the result notes named under Software and data availability.

Appendix B Supporting results

B.1 The exact sampler on M0

Coverage. We read coverage off the ranks of Fig. 9A, for central credible intervals at five levels from 50% to 99%. In 29 of 30 parameter–level pairs the coverage is inside the binomial 95% band. One or two misses over 30 pairs are expected by chance. The one miss is the decay rate at 99%, where $L = 100$ draws per chain give the coarsest ranks.

Table 7: **The runs behind the results.** Hours are from Slurm accounting. The mixed route's omit 274.1 lost to a fault, and the budget's are core-hours held.

Run	Slurm jobs	CPU-hours
Exact reference	18157561–65, 18271545, 18276511–15, 18420986–90	6,810
Exact calibration	18276794, 18628508, 18689602	33,900
Exact, production chain	18751635	507.5
Mixed route	19177653–19177825, 19238017–56, 19292223–32, 19334375–76, 19334378–80, 19411150	402.4
M0 scoring	19684097–101, 19827688	12,978
Reduced-model budget	20041038–40	1,978

A production-length chain matches the reference. One chain of the length we used in the calibration runs is not distinguishable from the reference on any marginal (Fig. 9B). The means differ by at most 0.12 reference standard deviations, and the ratio of standard deviations is between 0.91 and 1.19. This chain cost 507.5 CPU-hours. Its effective sample size is low, so this comparison has low resolving power. The production chain separates the FBA constant from its matched null by a small margin, 0.020 standard deviations.

B.2 The mixed route: holding S fixed

S is computed once, at a reference parameter value. A sensitivity check re-estimated it at the posterior mean on three datasets. The marginals moved detectably on 5 of 18 comparisons, each by at most 0.027 standard deviations over its floor. Against a matched null of two identical fits, two of the five are about twice the noise, one is 1.3 times it, and two are within it.

B.3 The mean-field swap: goodness check and limit

The goodness check. The check as first written fails a description known to be correct in about half of all cases: a correct description passes only 36–56% of the time. We therefore compare the check with its own null distribution. With δ, the mean field falls inside the null's 95% range at 26 of 28 targets, against 5 of 28 without δ.

The mean-field model is the limit of the hybrid's pinned path. The hybrid exchanges state every second. When the driver follows the mean-field path, this path converges to the one-ODE mean-field model as the step shrinks. That is a limit of the deterministic path, and not of the stochastic hybrid, whose gap is the subject of Section 3.3. Extrapolated to a zero step, the two agree to 0.0122 in log concentration at all five test parameter sets. One mean-field solve takes 1.1 s on M0 and 9–12 s on the full reduced model.

B.4 M0's crossed comparison

Table 8 gives the coverages and widths that Fig. 6 plots.

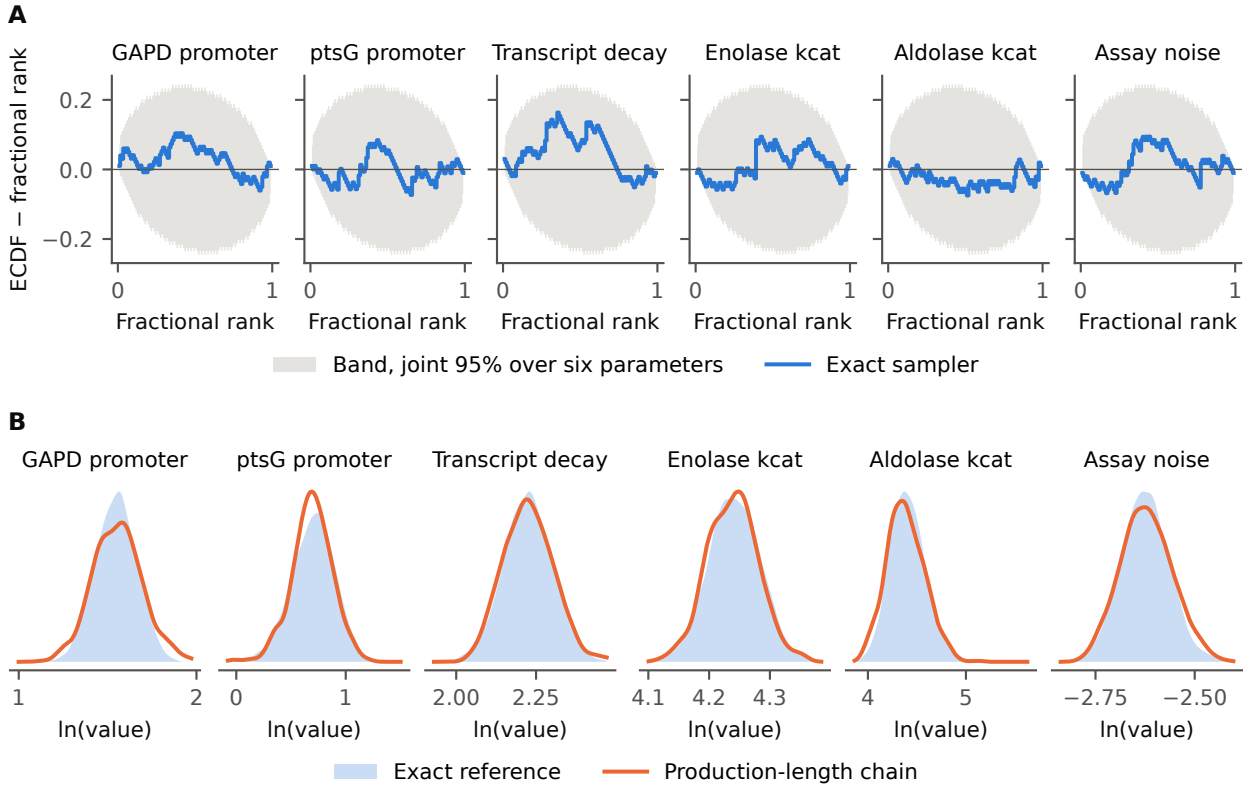


Figure 9: **The exact sampler on M0.** A: rank-ECDF of the truth over 50 prior-drawn datasets, against simultaneous bands at a joint level of 95%. The grey region is the band, and the line is the rank-ECDF minus the fractional rank. B: marginal posteriors of the natural log of each parameter on one dataset: the reference as a filled blue density, and a production-length chain as the orange line.

Table 8: **M0's crossed comparison. Scored** against the truth over 50 datasets, as percentage coverage ± one standard error. Width is the median ratio of the 90% interval to that of the exact posterior with the hybrid. The references at these sizes are 49% and 88.2%. The ΔQ columns do not meet the sizing rules in Section 3.5, so they are not clean calibration numbers.

		Q, lactate export			ΔQ, knockdown effect		
Posterior	Forward	50%	90%	Width	50%	90%	Width
Exact	Hybrid	62.0 ± 6.9	92.0 ± 3.8	1	62.0 ± 6.9	98.0 ± 2.0	1
Exact	Mean field	58.0 ± 7.0	94.0 ± 3.4	0.988	46.0 ± 7.0	90.0 ± 4.2	0.733
Exact	Mean field $+ \delta$	56.0 ± 7.0	90.0 ± 4.2	0.988	48.0 ± 7.1	88.0 ± 4.6	0.733
Mixed	Hybrid	50.0 ± 7.1	88.0 ± 4.6	0.823	52.0 ± 7.1	96.0 ± 2.8	0.964
Mixed	Mean field	54.0 ± 7.0	82.0 ± 5.4	0.772	32.0 ± 6.6	86.0 ± 4.9	0.635
Mixed	Mean field $+ \delta$	48.0 ± 7.1	84.0 ± 5.2	0.772	26.0 ± 6.2	78.0 ± 5.9	0.635